

Orforglipron compared with dapagliflozin in adults with type 2 diabetes and inadequate glycaemic control with metformin (ACHIEVE-2): a multicentre, randomised, non-inferiority, open-label, phase 3 trial



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Summary

Background Type 2 diabetes is a complex metabolic disorder often requiring combination therapy for optimal glycaemic control. This study assessed the efficacy and safety of orforglipron, an oral, non-peptide GLP-1 receptor agonist, versus dapagliflozin, an oral SGLT2 inhibitor, in participants with type 2 diabetes and inadequate glycaemic control with metformin.

Methods This 40-week, phase 3, multicentre, open-label (orforglipron dose-blinded), randomised study was conducted in 73 sites across six countries. The study included adults with type 2 diabetes using metformin (≥ 1500 mg/day) with glycated haemoglobin (HbA_{1c}) concentrations between 7·0% and 10·5% (53–91 mmol/mol), stable bodyweight ($\pm 5\%$), and a BMI of 23·0 kg/m² or more. Participants were randomly assigned in a 1:1:1:1 ratio to receive once-daily oral orforglipron (3 mg, 12 mg, or 36 mg) or dapagliflozin 10 mg. The primary endpoint was change from baseline in HbA_{1c} at week 40, using a non-inferiority margin of 0·3% for orforglipron versus dapagliflozin; analysis was based on the treatment-regimen estimand (data obtained regardless of study intervention discontinuation or initiation of additional glucose-lowering agents). The treatment regimen estimand was the primary estimand, with the efficacy estimand considered supportive. Safety was assessed in all randomised participants who received at least one dose of study drug. This trial is registered at ClinicalTrials.gov (NCT06192108, completed).

Findings From Jan 10, 2024, to Sept 26, 2025, 1404 adults were assessed and 962 participants were randomly assigned to orforglipron 3 mg (n=240), 12 mg (n=241), or 36 mg (n=241), or dapagliflozin 10 mg (n=240). The baseline population included 474 (49%) females and had a mean age of 56·1 years (SD 11·5), HbA_{1c} of 8·14% (1·04), type 2 diabetes duration of 8·0 years (6·7), and BMI of 32·6 kg/m² (6·6). At week 40, for the treatment regimen estimand, all orforglipron doses were non-inferior to dapagliflozin in reducing HbA_{1c}. Change from baseline in mean HbA_{1c} was –1·23% (SE 0·08), –1·50% (0·08), and –1·56% (0·09) with orforglipron 3 mg, 12 mg, and 36 mg, respectively, versus –0·81% (0·07) with dapagliflozin 10 mg; estimated treatment difference versus dapagliflozin was –0·42% (95% CI –0·62 to –0·23), –0·70% (–0·90 to –0·49), and –0·75% (–0·96 to –0·55) with orforglipron 3 mg, 12 mg, and 36 mg, respectively (all $p < 0·0001$). The most frequent adverse events associated with orforglipron were mild-to-moderate gastrointestinal events, observed in 112 (47%) of 240 participants receiving orforglipron 3 mg, 112 (46%) of 241 receiving 12 mg, and 130 (54%) of 241 receiving 36 mg, versus 29 (12%) of 240 receiving dapagliflozin. More study intervention discontinuations were observed with orforglipron 3 mg (35 [15%] of 240), 12 mg (44 [18%] of 241), and 36 mg (47 [20%] of 241) than with dapagliflozin (14 [6%] of 240). No severe episode of hypoglycaemia was reported.

Interpretation Orforglipron demonstrated superior glycaemic control compared with dapagliflozin, with a tolerability profile consistent with the GLP-1 receptor agonist class, including increased rates of discontinuation due to adverse events, positioning it as a potential effective oral treatment option for type 2 diabetes.

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Introduction

Type 2 diabetes is a complex metabolic disorder characterised by impaired glycaemic control as a result of increased insulin resistance and progressive β -cell

failure.¹ Type 2 diabetes is associated with obesity, hypertension, and dyslipidaemia.² The condition poses a substantial health burden due to both microvascular and macrovascular complications.³

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See Online for appendix

Research in context

Evidence before this study

Metformin might not result in optimal glycaemic control as a first-line monotherapy for type 2 diabetes. Other classes of glucose lowering medications including SGLT2 inhibitors and GLP-1 receptor agonists constitute adequate oral options for treatment intensification after metformin. SGLT2 inhibitors have well documented glucose-lowering efficacy along with cardiovascular and renal benefits. However, from a glycaemic efficacy perspective, many patients might not achieve glycaemic targets with SGLT2 inhibitor-based regimens, highlighting the need for additional effective oral treatment options.

Oral semaglutide, an oral formulation of a GLP-1 receptor agonist, demonstrated superior glycaemic control compared with empagliflozin, an SGLT2 inhibitor, in adults with type 2 diabetes inadequately controlled with metformin. However, oral semaglutide requires fasting conditions and water restrictions, which might not meet patient needs.

Orforglipron is a non-peptide oral GLP-1 receptor agonist approved for the treatment of overweight and obesity in the USA and under investigation for type 2 diabetes and other cardiometabolic diseases. Orforglipron can be taken anytime in the day without food or water restrictions. However, it is currently unknown how orforglipron compares with an SGLT2 inhibitor in people with type 2 diabetes inadequately controlled with metformin.

We searched PubMed and clinical study registries on Jan 14, 2026, using the terms “orforglipron”, “LY3502970”, “glucagon-like peptide-1 receptor agonist”, “dapagliflozin”, “sodium-glucose cotransporter 2 inhibitors”, and “type 2 diabetes” with no date or study duration restrictions, to identify any previous research on orforglipron and dapagliflozin. No head-to-head randomised clinical study comparing orforglipron with any SGLT2 inhibitors existed before the present study, ACHIEVE-2.

Added value of this study

ACHIEVE-2 is the first phase 3, randomised study to directly compare orforglipron with an SGLT2 inhibitor, dapagliflozin, in adults with type 2 diabetes inadequately controlled by metformin. In this study, once-daily oral administration of orforglipron without food or water restriction resulted in non-inferior, and superior improvements in glycaemic control compared with dapagliflozin, without increasing the risk of clinically significant hypoglycaemia. Compared with dapagliflozin, orforglipron 12 mg and 36 mg demonstrated greater bodyweight reduction after 40 weeks, and orforglipron 36 mg demonstrated improvement in cardiometabolic risk factors, including triglycerides, non-HDL cholesterol, and systolic blood pressure. Orforglipron also demonstrated a safety profile consistent with the GLP-1 receptor agonist class, mainly characterised by mild-to-moderate and generally transient gastrointestinal adverse events, as well as increased rates of discontinuation due to adverse events. The head-to-head results from the ACHIEVE-2 study provide direct comparative evidence that orforglipron is non-inferior and superior to dapagliflozin in the predefined endpoints.

Implications of all the available evidence

The improvements in glycaemic control, bodyweight reduction, and improvements in selected cardiometabolic risk factors observed with orforglipron compared with dapagliflozin in the present ACHIEVE-2 study, combined with findings from ACHIEVE-1 and ACHIEVE-3, suggest that orforglipron constitutes an important therapeutic option for people living with type 2 diabetes. Findings could inform decision making for clinicians selecting second-line therapy to combine with metformin for the treatment of type 2 diabetes.

Despite a variety of available pharmacotherapies, the proportion of people with type 2 diabetes who reach their glycaemic goals remains low.⁴ Metformin has traditionally been recommended as a first-line glucose-lowering agent for the management of type 2 diabetes, but it might not result in optimal glycaemic control as a monotherapy.⁵ As the global prevalence of type 2 diabetes continues to rise, there is a pressing need for novel glucose-lowering agents to achieve optimal glycaemic control and facilitate treatment initiation and adherence.

The American Diabetes Association/European Association for the Study of Diabetes guidelines recommend a patient-centred, comorbidity-driven approach to type 2 diabetes management.^{5,6} Medication selection should support the achievement of individualised therapeutic goals, including glycaemic control, weight reduction, and cardiovascular or renal

risk reduction.⁷ SGLT2 inhibitors and GLP-1 receptor agonists are both recommended for the management of patients with type 2 diabetes and comorbidities, alone or in combination with metformin, based on their demonstrated efficacy beyond glycaemic control.⁶

The choice of administration route can also impact disease management. For instance, patients might prefer oral formulations over injections.⁸ Whereas SGLT2 inhibitors are oral medications, most GLP-1 receptor agonists are currently administered via injection.

Orforglipron is a novel, chemically synthesised, non-peptide (small molecule) GLP-1 receptor agonist designed for daily oral administration without restrictions related to food or water intake.⁹ Orforglipron is currently approved for the treatment of overweight and obesity in the USA and is under investigation for type 2 diabetes and other cardiometabolic diseases. In

the phase 3 ACHIEVE-1 study, in adults with early type 2 diabetes inadequately controlled with diet and exercise, orforglipron significantly reduced glycated hemoglobin (HbA_{1c}) concentrations and bodyweight over 40 weeks.⁹ The ACHIEVE-3 study demonstrated orforglipron (12 mg and 36 mg) non-inferiority and superiority to oral semaglutide (7 mg and 14 mg) in HbA_{1c} and bodyweight reductions in people with type 2 diabetes inadequately controlled with metformin.¹⁰ Additionally, in the ATTAIN-2 study, orforglipron showed superior reduction in bodyweight compared with placebo in adults with overweight or obesity with type 2 diabetes, with a safety profile similar to other GLP-1 receptor agonists.¹¹ Dapagliflozin, a widely prescribed oral SGLT2 inhibitor, is associated with improved glycaemic control, modest weight reduction, and reduced cardiovascular risk in people with type 2 diabetes.^{12–14} Evidence from the phase 3 PIONEER-2 study showed that oral semaglutide, a GLP-1 receptor agonist, provides greater reductions in HbA_{1c} and bodyweight compared with another SGLT2 inhibitor, empagliflozin, in people with type 2 diabetes inadequately controlled with metformin.¹⁵ However, whether these findings extend to orforglipron requires direct evaluation.

The ACHIEVE-2 study compared the efficacy and safety of orforglipron (3 mg, 12 mg, and 36 mg) versus dapagliflozin 10 mg in people with type 2 diabetes and inadequate glycaemic control with metformin. The study assessed and compared the antihyperglycaemic effects of orforglipron versus dapagliflozin, as well as their effect on bodyweight, cardiometabolic parameters, and safety.

Methods

Study design and participants

This 40-week, phase 3, multicentre, multinational, randomised, open-label (orforglipron dose-blinded), active comparator-controlled, parallel-arm study was conducted in 73 sites across six countries (China, Germany, Mexico, Poland, Taiwan, and the USA). Eligible participants were 18 years or older with type 2 diabetes inadequately controlled using metformin (at least 1500 mg/day during the 90 days before screening and maintained through randomisation) with HbA_{1c} concentrations of 7·0% or more to 10·5% or less (≥ 53 to ≤ 91 mmol/mol), BMI 23·0 kg/m² or more, and with stable bodyweight (5% margin, for at least the 90 days before screening).

Key exclusion criteria included type 1 diabetes, severe hypoglycaemia in the last 6 months, diabetic retinopathy or macular oedema requiring treatment, history of pancreatitis, gastric emptying abnormality, liver disease (except metabolic dysfunction-associated steatotic liver disease [MASLD]), alanine aminotransferase (ALT) or aspartate aminotransferase (AST) concentration more than five times the upper limit of normal (ULN), and estimated glomerular filtration rate less than 45 mL/min

per 1·73 m² (calculated by CKD-EPI cystatin-c equation). Participants receiving strong CYP3A inhibitors or inducers, drugs that are P-glycoprotein or breast cancer resistance protein substrates with narrow therapeutic index, or strong organic anion-transporting polypeptide inhibitors were also excluded. A full list of eligibility criteria is provided in the appendix (pp 10–13).

The protocol was approved by local institutional review boards, and the study was conducted in accordance with the Declaration of Helsinki guidelines on good clinical practices, the Council for International Organisations of Medical Sciences International Ethical Guidelines, and the International Conference on Harmonisation Good Clinical Practice Guideline. All study participants provided written informed consent before participating in the study. A full list of institutional review boards is provided in the appendix (pp 7–9). The clinical trial protocol (pp 59–191) and the statistical analysis plan (pp 192–251) are provided in the appendix. This trial is registered at ClinicalTrials.gov (NCT06192108, completed).

Randomisation and masking

Participants were randomly assigned in a 1:1:1 ratio to receive orforglipron (3 mg, 12 mg, 36 mg) or dapagliflozin 10 mg. This publication presents data from the investigational orforglipron capsule formulations of 3 mg, 12 mg, and 36 mg. The doses have been shown as the equivalent to tablet doses of 2·5 mg, 9 mg, and 17·2 mg, respectively¹⁶ (appendix p 46).

Randomisation was conducted using a computer-based interactive web-response system (IWRS) and stratified by country and HbA_{1c} ($\leq 8\cdot0\%$ or $>8\cdot0\%$ [64 mmol/mol]). This open-label study compared orforglipron and dapagliflozin 10 mg, with orforglipron doses being double-blinded. As all participants escalated through the first two doses of orforglipron, the actual dose could be inferred during the first 8 weeks, before the divergence of the dose escalation scheme (appendix p 46). Investigators, site staff, clinical monitors, and participants remained masked to the assigned maintenance dose of orforglipron (ie, 3 mg, 12 mg, or 36 mg) until the study was complete, whereas treatment assignment to dapagliflozin 10 mg was not blinded due to difference in formulation. Emergency unblinding was performed through the IWRS only if the participant's wellbeing required knowledge of the participant's orforglipron dose assignment. All unblinding events were documented and reported by the IWRS.

In this study, no dedicated data monitoring committee was employed. A program-level committee regularly reviewed unblinded safety data from ACHIEVE-4 (NCT05803421), the longest orforglipron study in people with type 2 diabetes and obesity or overweight at increased cardiovascular risk. In the event that a need was identified during routine blinded safety monitoring by the sponsor, the program-level committee could

assess data from other studies. No such event occurred during ACHIEVE-2 or other studies.

Procedures

The study period included up to 4 weeks of screening and lead-in period, a 40-week study intervention period, and 2 weeks of safety follow-up (appendix p 46). Orforglipron or dapagliflozin were administered orally once daily. There were no restrictions on the time of day each dose was taken; however, it was recommended to take the dose at approximately the same time each day. Participants receiving orforglipron initiated with a 1 mg dose, escalating to 3 mg at week 4, then followed by no more than doubling of dose at 4-week intervals until the randomised dose of 3 mg, 12 mg, or 36 mg was reached. Participants assigned to dapagliflozin were initiated at 10 mg and maintained this dose for the remainder of the study intervention period (appendix p 46). Further details on study design and procedures are described in the appendix (pp 14–21) and in the study protocol (pp 59–191).

Initiation of new glucose-lowering medication was allowed according to specific rescue criteria: (1) for protocol-defined severe persistent hyperglycaemia; (2) in participants who permanently discontinue the assigned study intervention but remain in the study; or (3) during the safety follow-up period. Orforglipron dose de-escalation was not allowed. In case of intolerable gastrointestinal symptoms, participants were advised to adapt meals, use symptomatic medications and, if needed, allowed to temporarily interrupt study drug for up to two consecutive doses of study treatment. If participants missed three to six consecutive doses, orforglipron treatment was restarted at a 3 mg dose and re-escalation or continuation per protocol was instructed at investigator's discretion and based on when the interruption occurred during the study. If participants missed seven consecutive doses or more, they re-started treatment at 3 mg and re-escalated to the assigned randomised dose. Further details on the initiation of new glucose-lowering medication and instructions on dose re-escalation are described in the appendix (pp 14–16).

The sponsor used a central laboratory to analyse urine and blood samples taken for efficacy and safety analysis.

Outcomes

The primary endpoint was change from baseline in HbA_{1c} at week 40, assessing non-inferiority (margin of 0·3%) of orforglipron 3 mg, 12 mg, and 36 mg to dapagliflozin. Key secondary endpoints, all assessing superiority of orforglipron to dapagliflozin, were change from baseline in HbA_{1c} at week 40 (all doses), proportion of participants reaching HbA_{1c} targets of less than 7·0% (53 mmol/mol) and less than or equal to 6·5% (48 mmol/mol) at week 40 (all doses), change and percent change from baseline in mean bodyweight at week 40

(12 mg and 36 mg doses), change from baseline in mean systolic blood pressure (SBP) at week 40 (36 mg dose), and percent change from baseline in mean non-HDL cholesterol and triglycerides at week 40 (36 mg dose). The above-described primary endpoint and all the key secondary endpoints were adjusted for multiplicity for the described orforglipron doses (appendix pp 23–24, 47). Other secondary efficacy endpoints included change and percent change from baseline in bodyweight (3 mg dose); change in triglycerides, non-HDL cholesterol, and SBP (3 mg and 12 mg doses); the proportion of participants reaching an HbA_{1c} target of less than 5·7% (39 mmol/mol; all doses); proportions of participants reaching bodyweight reduction targets of 5% or greater, 10% or greater, and 15% or greater (all doses); and change from baseline in fasting serum glucose (FSG), daily mean 7-point self-monitored blood glucose (SMBG) profiles, BMI, waist circumference, diastolic blood pressure (DBP), total cholesterol, HDL cholesterol, LDL cholesterol, VLDL cholesterol, and patient-reported outcomes of Short Form-36 Health Survey Version 2 (SF-36 v2) and Diabetes Treatment Satisfaction Questionnaire-change (DTSQc; all doses). A summary of primary and secondary endpoints can be found in the appendix (pp 23–24). Exploratory endpoints included subgroup analyses of model-based estimate (MBE) mean HbA_{1c} change from baseline for the treatment regimen estimand. Further details on study endpoints can be found in the study protocol (appendix pp 59–191).

Safety endpoints included serious adverse events (SAEs), treatment-emergent adverse events (TEAEs), and discontinuations due to adverse events. Adverse events of special interest included hypoglycaemia, severe hypoglycaemia, initiation of rescue therapy for severe persistent hyperglycaemia, adjudicated pancreatitis, gallbladder and biliary tract disorders, diabetic retinopathy complications, major adverse cardiovascular events (MACE), and urinary tract infections. Ophthalmological evaluation of fundus photographs for diabetic retinopathy or macular oedema was done for all participants at baseline and week 40, with an additional assessment at week 24 for participants found to have any degree of diabetic retinopathy or macular oedema at baseline. Further details of fundus photography assessments are provided in the study protocol. During the trial, medical monitoring of blinded participant data was used to ensure participant safety, including reviews of laboratory alerts, adverse events or SAEs, and periodic medical data listings.

Statistical analysis

The statistical analysis methods were prespecified in the statistical analysis plan (appendix pp 192–251). A sample size of 888 participants was calculated to ensure at least 90% power in assessing non-inferiority in the primary endpoint of HbA_{1c} with a 0·3% non-inferiority margin for orforglipron 3 mg, 12 mg, and 36 mg, each tested in

parallel, against dapagliflozin 10 mg at a two-sided alpha level of 0·025. The sample size calculation assumed a treatment effect of $-0\cdot1\%$ for the mean change in HbA_{1c} from baseline to week 40 between orforglipron groups and dapagliflozin, and a common SD of $1\cdot2\%$. The power calculations were performed with the software nQuery Version 9.1.0.0.

Efficacy analyses were conducted for all randomised participants. Two estimands were predefined to assess the study's primary and key secondary efficacy objectives: the treatment regimen estimand and the efficacy estimand. The treatment regimen estimand was the primary estimand, with the efficacy estimand considered supportive.

The treatment regimen estimand included all datapoints obtained during the treatment period defined

as at or after baseline and up to the last visit within the treatment period, regardless of study intervention discontinuation or initiation of additional glucose-lowering agents (any antihyperglycaemic medications that are used for more than 14 days). Continuous outcomes were analysed using an ANCOVA model and binary outcomes were analysed using a logistic regression model, with missing data imputed 50 times. Models were adjusted for treatment group, country, baseline HbA_{1c} ($\leq 8\cdot0\%$, $> 8\cdot0\%$), baseline parameter value, and interactions of treatment-by-baseline HbA_{1c} ($\leq 8\cdot0\%$, $> 8\cdot0\%$) and treatment-by-baseline parameter. These models incorporated imputed data for missing endpoints at week 40, as detailed in the statistical analysis plan.

The efficacy estimand estimated the treatment effect between orforglipron and dapagliflozin if participants

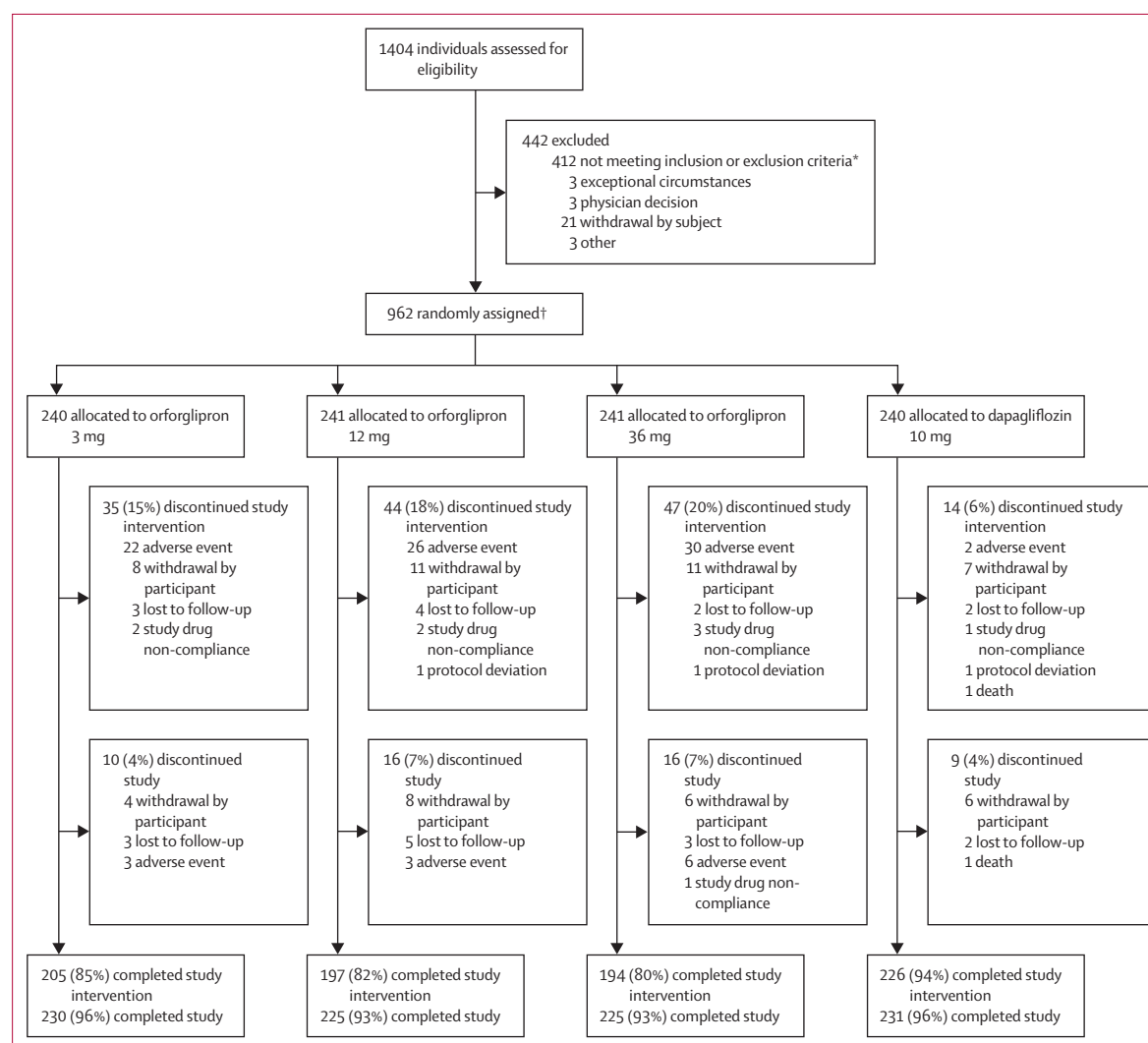


Figure 1: Trial profile

Study completion is defined as completing both the 40-week treatment period and the safety follow-up period. If participants discontinue the study during the treatment period, they also discontinue the study intervention, either earlier, or at the same time. Reason for study discontinuation is considered the earliest discontinuation reason from either treatment period or safety follow-up period. *For full details on the inclusion criteria and the complete list refer to the appendix (pp 10–13). †All randomised participants received study treatment.

had remained on their randomly assigned study intervention for 40 weeks and would not have initiated additional glucose-lowering agents. The efficacy estimand was estimated using a mixed model for repeated measures (MMRM), based upon data observed at or after baseline and up to the earliest intercurrent

event of discontinuation or initiation of additional glucose-lowering agents (used for more than 14 days). The MMRM included variables for treatment group, visit, country, baseline HbA_{1c} ($\leq 8.0\%$, $> 8.0\%$), and visit-by-treatment-by-baseline HbA_{1c} ($\leq 8.0\%$, $> 8.0\%$) interaction and visit-by-treatment-by-baseline parameter

	Orforglipron 3 mg (n=240)	Orforglipron 12 mg (n=241)	Orforglipron 36 mg (n=241)	Dapagliflozin 10 mg (n=240)	Overall population (n=962)
Demographic variables					
Age, years	55.1 (11.8)	55.9 (11.6)	56.6 (11.5)	56.6 (11.1)	56.1 (11.5)
Sex					
Female	117 (49%)	121 (50%)	123 (51%)	113 (47%)	474 (49%)
Male	123 (51%)	120 (50%)	118 (49%)	127 (53%)	488 (51%)
Race*					
White	119 (50%)	125 (52%)	131 (54%)	125 (52%)	500 (52%)
American Indian or Alaska Native	67 (28%)	62 (26%)	53 (22%)	61 (25%)	243 (25%)
Asian	43 (18%)	42 (17%)	44 (18%)	42 (18%)	171 (18%)
Multiple	5 (2%)	5 (2%)	11 (5%)	6 (3%)	27 (3%)
Black or African American	6 (3%)	6 (2%)	2 (1%)	6 (3%)	20 (2%)
Ethnicity					
Not Hispanic or Latino	124 (52%)	125 (52%)	122 (51%)	124 (52%)	495 (51%)
Hispanic or Latino	114 (48%)	114 (47%)	116 (48%)	114 (48%)	458 (48%)
Not reported	2 (1%)	2 (1%)	3 (1%)	2 (1%)	9 (1%)
Country					
Mexico	99 (41%)	100 (41%)	100 (41%)	100 (42%)	399 (41%)
Poland	36 (15%)	36 (15%)	36 (15%)	35 (15%)	143 (15%)
USA	35 (15%)	35 (15%)	36 (15%)	35 (15%)	141 (15%)
Germany	30 (13%)	29 (12%)	29 (12%)	29 (12%)	117 (12%)
China	25 (10%)	26 (11%)	25 (10%)	26 (11%)	102 (11%)
Taiwan	15 (6%)	15 (6%)	15 (6%)	15 (6%)	60 (6%)
Clinical variables					
HbA _{1c} , %	8.17 (1.08)	8.11 (1.05)	8.18 (1.08)	8.09 (0.98)	8.14 (1.04)
HbA _{1c} , mmol/mol	65.8 (11.8)	65.2 (11.5)	65.9 (11.8)	65.0 (10.7)	65.5 (11.4)
HbA _{1c} category					
HbA _{1c} $\leq 8.0\%$	126 (53%)	129 (54%)	128 (53%)	130 (54%)	513 (53%)
HbA _{1c} $> 8.0\%$	114 (48%)	112 (46%)	113 (47%)	110 (46%)	449 (47%)
Fasting serum glucose, mg/dL	159.4 (52.5)	160.1 (51.1)	157.0 (49.5)	152.3 (49.4)	157.2 (50.6)
Fasting serum glucose, mmol/L	8.9 (2.9)	8.9 (2.8)	8.7 (2.8)	8.5 (2.7)	8.7 (2.8)
Bodyweight, kg	90.1 (20.4)	92.1 (23.6)	88.1 (22.0)	89.4 (21.1)	89.9 (21.8)
BMI, kg/m ²	32.4 (6.5)	33.2 (7.0)	32.2 (6.6)	32.4 (6.5)	32.6 (6.6)
Waist circumference, cm	107.0 (14.8)	108.0 (15.7)	106.3 (14.7)	107.2 (14.8)	107.1 (15.0)
Duration of type 2 diabetes, years	7.4 (6.4)	7.9 (6.6)	8.1 (7.1)	8.5 (6.7)	8.0 (6.7)
Previous use of GLP-1 receptor agonist (yes)	15 (6%)	12 (5%)	12 (5%)	15 (6%)	54 (6%)
Blood pressure, mm Hg					
Systolic	127.6 (11.4)	128.1 (14.2)	127.7 (12.8)	128.3 (12.4)	127.9 (12.7)
Diastolic	79.5 (8.2)	80.2 (8.3)	79.5 (9.2)	80.0 (8.0)	79.8 (8.4)
Pulse rate, beats per min	74.1 (10.2)	75.2 (9.7)	74.7 (10.2)	73.8 (10.0)	74.5 (10.0)
eGFR, mL/min per 1.73 m ²	87.8 (20.4)	87.5 (20.1)	87.8 (18.8)	85.8 (20.7)	87.2 (20.0)
Triglycerides, mg/dL	167.9 (106.4)	184.9 (221.3)	174.0 (131.5)	162.7 (78.7)	172.4 (144.8)
Non-HDL cholesterol, mg/dL	126.3 (44.1)	131.6 (43.1)	133.7 (52.7)	128.7 (41.6)	130.1 (45.6)
Data are n (%) or mean (SD). eGFR=estimated glomerular filtration rate. HbA _{1c} =glycated haemoglobin. *One participant missing for race parameter, n=240 for orforglipron 12 mg.					
Table 1: Baseline demographics and clinical variables of all randomly assigned participants					

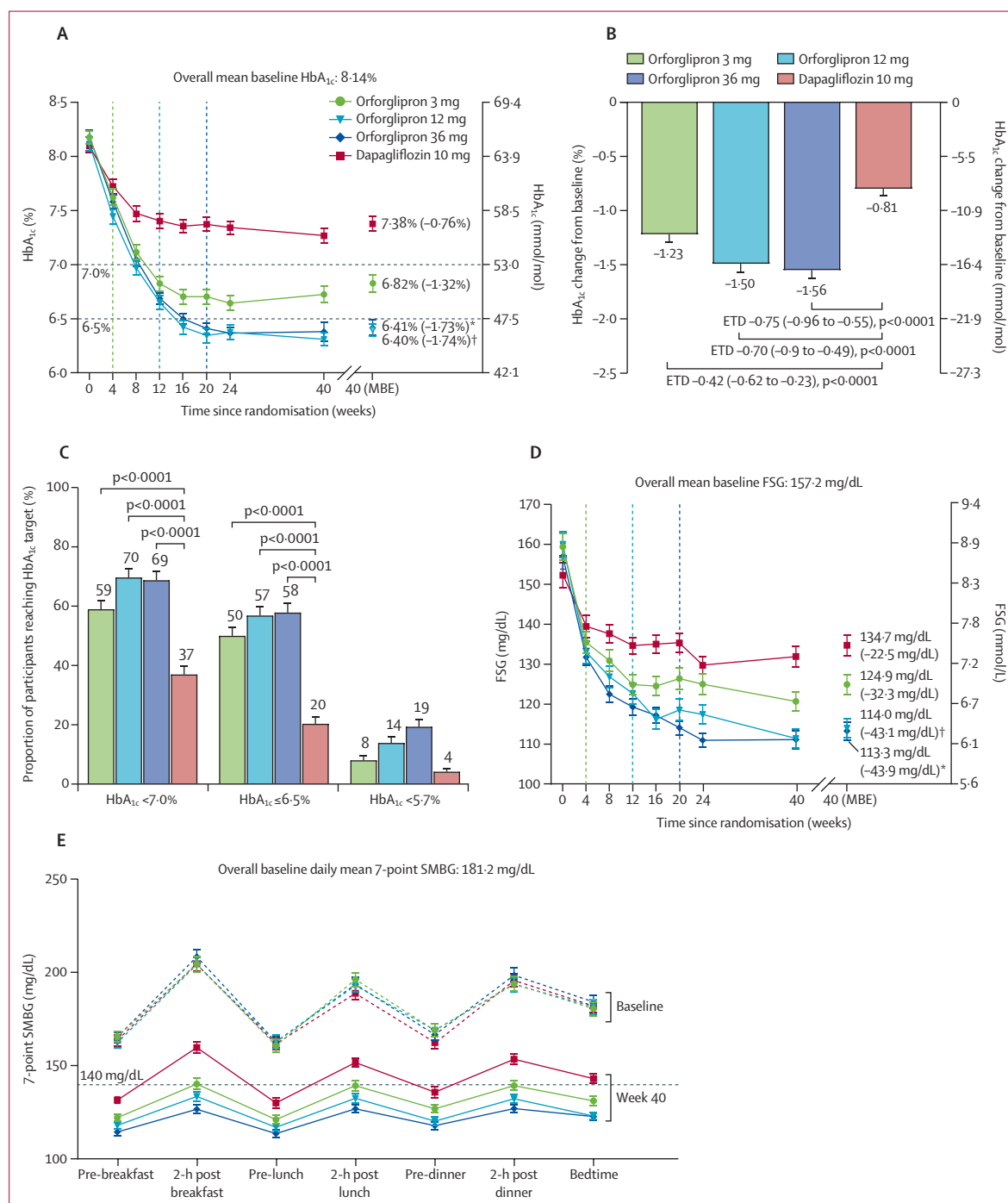


Figure 2: HbA_{1c}, proportion of participants reaching HbA_{1c} targets, FSG, and 7-point SMBG profiles

(A) Over time curve data are observed mean HbA_{1c} for the efficacy estimand; MBE at week 40 is actual value (change from baseline) for the efficacy estimand. (B) Change from baseline in HbA_{1c} and ETD with p values versus dapagliflozin 10 mg at week 40 from ANCOVA for the treatment regimen estimand. (C) Proportion of participants achieving HbA_{1c} targets of <7.0%, ≤6.5%, and <5.7% and p values versus dapagliflozin 10 mg at week 40 from logistic regression analysis for the treatment regimen estimand. (D) Over time curve data are observed mean FSG for the efficacy estimand; MBE at week 40 is absolute value (change from baseline) for the efficacy estimand. (E) MBE mean 7-point SMBG profiles at baseline (dotted lines) and week 40 (full lines) for the efficacy estimand. In every panel, error bars denote SE. In panels A and D, vertical dotted lines denote the end of dose escalation periods for each orforglipron treatment group. In panels B and C, only p values tested for multiplicity (family-wise type-1 error) are shown. ETD=estimated treatment difference. FSG=fasting serum glucose. HbA_{1c}=glycated haemoglobin. MBE=model-based estimate. SMBG=self-monitored blood glucose. *Denotes orforglipron 36 mg group where lines overlap. †Denotes orforglipron 12 mg group where lines overlap.

interaction. For MMRM analysis, only data collected before the occurrence of any intercurrent events were used. Through the MMRM analysis, the potential efficacy measures (after the intercurrent events) were implicitly imputed as if participants did not experience intercurrent events. Binary outcomes were analysed using the same logistic regression model as treatment regimen estimand but with missing-at-random imputation.

A graphical testing procedure was applied under each estimand for key primary and secondary endpoints to strongly control the overall family-wise type I error rate at a two-sided alpha level of 0.05 (appendix pp 23–24, 47). The detailed methods on estimand imputations can be found in the appendix (pp 18–21) and statistical analysis plan (appendix).

Safety analyses were conducted on the safety population, comprising all participants who received at least one dose of study drug. The safety datapoints set included all datapoints obtained during the treatment period and the safety follow-up period defined as at or after baseline and up to the date of study withdrawal including the safety follow-up period regardless of study intervention discontinuation or initiation of additional glucose-lowering agents (any antihyperglycaemic medications that are used for more than 14 days). Assessment of hypoglycaemia excluded events that occurred after initiation of any new glucose-lowering agent.

R version 4.4.2 and SAS version 9.4 were used for statistical calculations. Additional details regarding the statistical analysis are provided in the appendix (pp 18–21).

Role of the funding source

The sponsor (Eli Lilly and Company) was involved in study design, data collection, data analysis, data interpretation, writing of the report, and the decision to submit for publication.

Results

Between Jan 10, 2024, and Sept 26, 2025, 1404 participants were assessed for eligibility and 962 were randomly assigned to orforglipron 3 mg (n=240), orforglipron 12 mg (n=241), orforglipron 36 mg (n=241), or dapagliflozin 10 mg (n=240; figure 1). A total of 911 participants (95%) completed the study, and 822 participants (85%) completed 40 weeks of treatment (study intervention). More study discontinuations were observed with orforglipron (3 mg: ten [4%] of 240; 12 mg: 16 [7%] of 241; 36 mg: 16 [7%] of 241) than with dapagliflozin (nine [4%] of 240), mainly due to withdrawal by participants across treatment groups. More study intervention discontinuations were observed with orforglipron 3 mg (35 [15%] of 240), 12 mg (44 [18%] of 241), and 36 mg (47 [20%] of 241) than with dapagliflozin 10 mg (14 [6%] of 240). Study intervention discontinuations due to adverse events occurred in 22 (9%) of

240 participants treated with orforglipron 3 mg, 26 (11%) of 241 treated with orforglipron 12 mg, and 30 (12%) of 241 treated with orforglipron 36 mg, and in two (1%) of 240 treated with dapagliflozin. Study intervention discontinuations due to participant withdrawal occurred in eight (3%) of 240, 11 (5%) of 241, and 11 (5%) of 241 participants treated with orforglipron 3 mg, 12 mg, and 36 mg, respectively, and in seven (3%) of 240 treated with dapagliflozin (figure 1).

Participant baseline demographics and clinical characteristics were similar between treatment groups (table 1). At baseline, the mean age of participants was 56.1 years (SD 11.5), and 474 (49%) of 962 were female. The mean duration of diabetes was 8.0 years (6.7), mean HbA_{1c} was 8.14% (1.04), mean bodyweight was 89.9 kg (21.8), and mean BMI was 32.6 kg/m² (6.6).

The ACHIEVE-2 study met all prespecified and multiplicity-controlled efficacy endpoints including non-inferiority and superiority of orforglipron doses (all doses) versus dapagliflozin 10 mg for change from baseline in HbA_{1c} at week 40, as well as superiority of orforglipron versus dapagliflozin in reaching HbA_{1c} targets of less than 7.0% and less than or equal to 6.5% (all doses), and in changes from baseline in bodyweight (12 mg and 36 mg), triglycerides, non-HDL cholesterol, and SBP (36 mg) at week 40. Data are presented for the efficacy estimand or the treatment regimen estimand as specified.

Participants treated with orforglipron 3 mg, 12 mg, and 36 mg experienced greater mean improvements in HbA_{1c} than those treated with dapagliflozin 10 mg (efficacy estimand; figure 2A, appendix p 48). Separation between orforglipron groups and dapagliflozin for observed mean HbA_{1c} was apparent from week 4 and maintained for the rest of the 40-week treatment period. All orforglipron doses had an MBE mean HbA_{1c} below 7% at week 40; mean values were 6.82% (SE 0.07), 6.40% (0.06), and 6.41% (0.08) with orforglipron 3 mg, 12 mg and 36 mg, respectively, compared with 7.38% (0.08) with dapagliflozin 10 mg (efficacy estimand; figure 2A). At week 40, mean change from baseline in HbA_{1c} was –1.23% (SE 0.08), –1.50% (0.08), and –1.56% (0.09) with orforglipron 3 mg, 12 mg, and 36 mg, respectively, compared with –0.81% (0.07) with dapagliflozin 10 mg (treatment regimen estimand; figure 2B, appendix pp 25–28). All orforglipron doses were non-inferior and superior to dapagliflozin 10 mg for mean HbA_{1c} reduction. Mean estimated treatment difference (ETD) versus dapagliflozin 10 mg was –0.42% (95% CI –0.62 to –0.23), –0.70% (–0.90 to –0.49), and –0.75% (–0.96 to –0.55) with orforglipron 3 mg, 12 mg, and 36 mg, respectively (treatment regimen estimand; all p<0.0001 versus dapagliflozin, multiplicity-controlled; figure 2B, appendix pp 25–28). Similar results and inferences were observed with the efficacy estimand datapoint set (appendix pp 29–33, 48). Exploratory subgroup analyses of MBE mean HbA_{1c} change from

baseline are shown in the appendix (pp 49–52) for the treatment regimen estimand.

At week 40, orforglipron treatment groups had significantly greater proportions of participants reaching

the prespecified HbA_{1c} targets. The target HbA_{1c} of less than 7.0% was reached by 140 (59%) of 240 participants receiving orforglipron 3 mg, 169 (70%) of 241 receiving orforglipron 16 mg, and 165 (69%) of 241 receiving

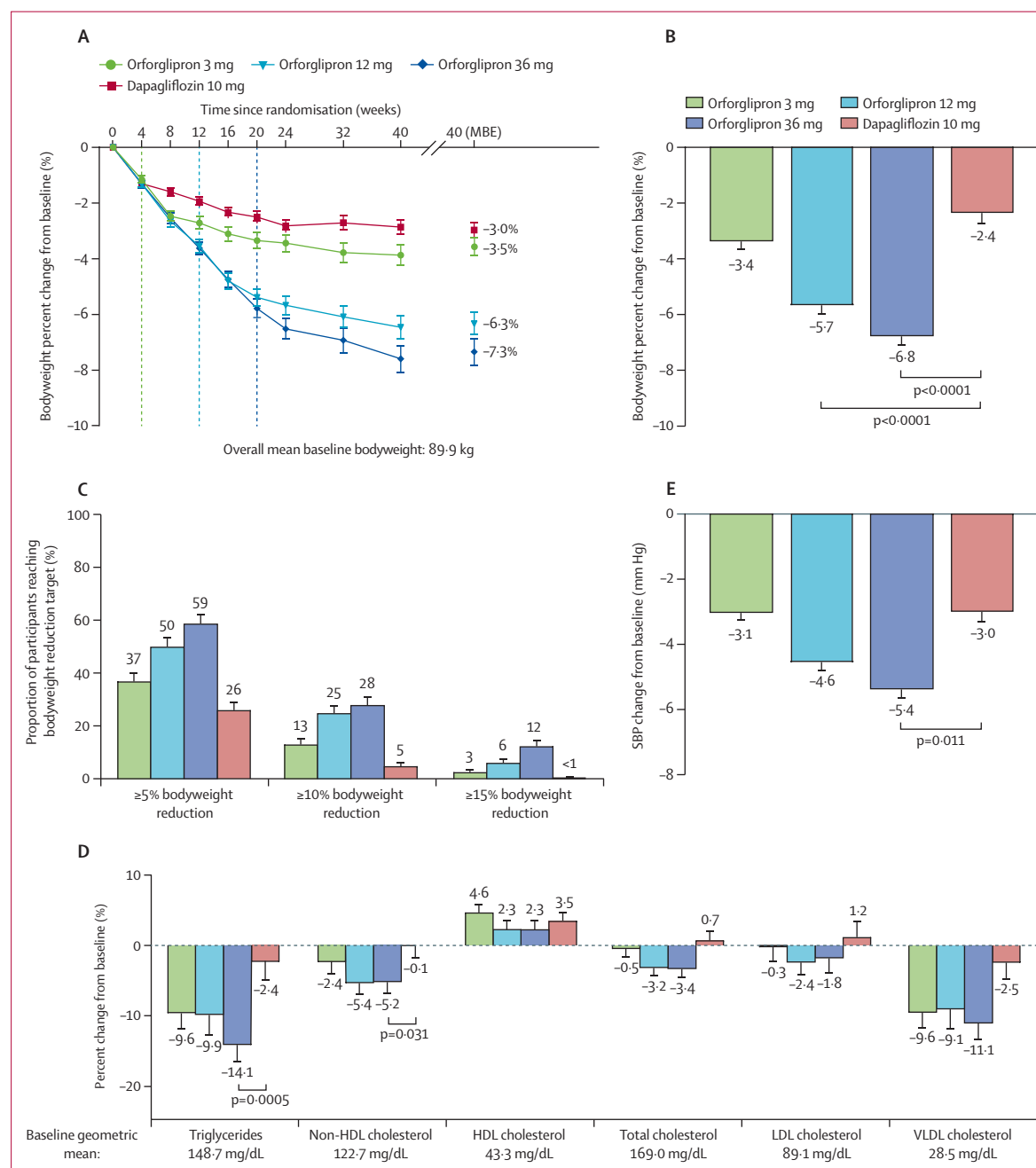


Figure 3: Bodyweight change, proportion of participants reaching bodyweight reduction targets, lipid profile, and SBP

(A) Over time curve data are observed percent change from baseline in bodyweight for the efficacy estimand; MBE at week 40 is actual value (change from baseline) for the efficacy estimand. (B) Percent change from baseline in bodyweight and p values versus dapagliflozin 10 mg at week 40 from ANCOVA for the treatment regimen estimand. (C) Proportion of participants reaching bodyweight reduction targets of ≥5%, ≥10%, and ≥15% at week 40 from logistic regression analysis for the treatment regimen estimand. (D) Percent change from baseline in lipids and p values versus dapagliflozin 10 mg at week 40 from ANCOVA for the treatment regimen estimand. (E) Change from baseline in SBP and p values versus dapagliflozin 10 mg at week 40 from ANCOVA for the treatment regimen estimand. In every panel, error bars denote SE. In panel A, vertical dotted lines denote the end of dose escalation periods for each orforglipron treatment group. In panels B, D, and E, only p values tested for multiplicity (family-wise type-1 error) are shown. MBE=model-based estimate. SBP=systolic blood pressure.

orforglipron 36 mg, respectively, compared with 90 (37%) of 240 receiving dapagliflozin 10 mg; risk difference versus dapagliflozin 22.0% (95% CI 13.5–30.5), 32.8% (24.4–41.3), and 31.8% (23.1–40.6) with orforglipron 3 mg, 12 mg, and 36 mg, respectively (treatment regimen estimand; all $p < 0.0001$ versus dapagliflozin, multiplicity-controlled; figure 2C, appendix pp 25–28). The target HbA_{1c} of less than or equal to 6.5% was reached by 119 (50%) of 240 participants receiving orforglipron 3 mg, 138 (57%) of 241 receiving orforglipron 16 mg, and 139 (58%) of 241 receiving orforglipron 36 mg, respectively, compared with 50 (20%) of 240 receiving dapagliflozin 10 mg; risk difference versus dapagliflozin 29.8% (95% CI 21.8–37.7), 36.6% (28.7–44.5), and 37.6% (29.4–45.8) with orforglipron 3 mg, 12 mg, and 36 mg, respectively (treatment regimen estimand; all $p < 0.0001$ versus dapagliflozin, multiplicity-controlled; figure 2C, appendix pp 25–28). Finally, a greater proportion of participants reached the HbA_{1c} target of less than 5.7% with orforglipron than with dapagliflozin 10 mg (treatment regimen estimand; figure 2C, appendix pp 25–28). Similar results and inferences were observed with the efficacy estimand datapoint set (appendix pp 29–33).

Participants treated with orforglipron experienced greater reductions in observed mean FSG throughout the study intervention period than participants treated with dapagliflozin 10 mg (efficacy estimand; figure 2D), as well as greater mean change from baseline at week 40 (efficacy estimand; appendix pp 29–33, 48). Compared with dapagliflozin 10 mg, participants treated with orforglipron had lower mean pre-meal and mean 2-h post-meal SMBG concentrations, with all orforglipron groups maintaining mean glucose concentrations at or below 140 mg/dL (7.8 mmol/L)⁷ at every timepoint (efficacy estimand; figure 2E). Greater changes from baseline in daily mean SMBG concentrations were observed with orforglipron compared with dapagliflozin (efficacy estimand; appendix pp 29–33). Treatment regimen estimand data for FSG and daily mean SMBG are presented in the appendix (pp 25–28).

Participants in orforglipron 12 mg and 36 mg treatment groups experienced greater observed mean bodyweight reduction from baseline than with dapagliflozin 10 mg. Weight reduction gradually increased with orforglipron 12 mg and 36 mg, indicating a dose-dependent effect, which did not appear to plateau by week 40 (efficacy estimand; figure 3A, appendix p 53). At week 40, mean percent change from baseline in bodyweight was –5.7% (–5.2 kg) and –6.8% (–6.3 kg) with orforglipron 12 mg and 36 mg, respectively, compared with –2.4% (–2.2 kg) with dapagliflozin 10 mg (treatment regimen estimand; all $p < 0.0001$ versus baseline; figure 3B, appendix pp 25–28, 53). Orforglipron 12 mg and 36 mg were superior to dapagliflozin 10 mg for mean percent change from baseline in bodyweight; ETD versus dapagliflozin 10 mg

was –3.3% (95% CI –4.4 to –2.2) and –4.4% (–5.6 to –3.2) with orforglipron 12 mg and 36 mg, respectively (treatment regimen estimand; $p < 0.0001$ versus dapagliflozin, multiplicity-controlled; figure 3B, appendix pp 25–28). Similar results and inferences were observed with the efficacy estimand datapoint set (appendix pp 29–33, 54). Bodyweight change from baseline with orforglipron 3 mg is presented in figure 3B and the appendix (pp 25–28) for the treatment regimen estimand, and in the appendix (pp 29–33, 54) for the efficacy estimand. Participants treated with orforglipron experienced numerically greater mean changes from baseline in BMI and waist circumference than with dapagliflozin (efficacy estimand; appendix pp 29–33, 54).

At week 40, compared with dapagliflozin, a greater proportion of participants treated with orforglipron reached bodyweight reduction targets of 5% or greater (orforglipron: 89 [37%] of 240 with 3 mg to 142 [59%] of 241 with 36 mg; dapagliflozin: 62 [26%] of 240), 10% or greater (orforglipron: 31 [13%] with 3 mg to 67 [28%] with 36 mg; dapagliflozin: 12 [5%]), and 15% or greater (orforglipron: six [3%] with 3 mg to 30 [12%] with 36 mg; dapagliflozin: one [$<1\%$]; treatment regimen estimand; figure 3C, appendix pp 25–28). Similar results were observed for the efficacy estimand (appendix pp 29–33).

At week 40, mean percent change from baseline in triglycerides was significantly greater with orforglipron 36 mg (–14.1%) than with dapagliflozin (–2.4%), with an ETD of –12.0% (95% CI –18.2 to –5.4; $p = 0.0005$ versus dapagliflozin, multiplicity-controlled, treatment regimen estimand; figure 3D, appendix pp 25–28). Mean percent change from baseline in non-HDL cholesterol was significantly greater with orforglipron 36 mg (–5.2%) than with dapagliflozin (–0.1%), with an ETD of –5.1% (–9.5 to –0.5; $p = 0.031$ versus dapagliflozin, multiplicity-controlled, treatment regimen estimand). Similar results and inferences were observed with the efficacy estimand datapoint set (appendix pp 29–33). Percent change from baseline in triglycerides and non-HDL cholesterol with orforglipron 3 mg and 12 mg, and in HDL cholesterol, total cholesterol, LDL cholesterol, and VLDL cholesterol for all doses are presented in figure 3D and the appendix (pp 25–28) for the treatment regimen estimand, and in the appendix (pp 29–33) for the efficacy estimand.

At week 40, participants treated with orforglipron 36 mg experienced significantly greater mean change from baseline in SBP (–5.4 mm Hg) than with dapagliflozin (–3.0 mm Hg) with an ETD of –2.4 mm Hg (95% CI –4.2 to –0.5; $p = 0.011$ versus dapagliflozin, multiplicity-controlled, treatment regimen estimand; figure 3E, appendix pp 25–28). Similar results and inferences were observed with the efficacy estimand datapoint set (appendix pp 29–33, 54). Changes from baseline in SBP with orforglipron 3 mg and 12 mg, and

	Orforglipron 3 mg (n=240)	Orforglipron 12 mg (n=241)	Orforglipron 36 mg (n=241)	Dapagliflozin 10 mg (n=240)
Participants with ≥ 1 TEAE	165 (69%)	168 (70%)	170 (71%)	129 (54%)
Serious adverse events	8 (3%)	14 (6%)	9 (4%)	11 (5%)
Adverse events leading to death*	0	0	0	1 (<1%)
Adverse events leading to study discontinuation*	3 (1%)	3 (1%)	6 (2%)	1 (<1%)
Adverse events leading to intervention discontinuation*	22 (9%)	26 (11%)	30 (12%)	3 (1%)
Gastrointestinal adverse events	14 (6%)	18 (7%)	20 (8%)	1 (<1%)
Vomiting	2 (1%)	5 (2%)	7 (3%)	0
Nausea	2 (1%)	3 (1%)	4 (2%)	0
Diarrhoea	3 (1%)	3 (1%)	0	0
Abdominal pain	1 (<1%)	1 (<1%)	2 (1%)	0
Dyspepsia	0	2 (1%)	2 (1%)	0
Other gastrointestinal adverse events	6 (3%)	4 (2%)	5 (2%)	1 (<1%)
Non-gastrointestinal adverse events	8 (3%)	8 (3%)	10 (4%)	2 (1%)
Adverse events occurring in $\geq 5\%$ of participants in any treatment group				
Nausea	47 (20%)	41 (17%)	53 (22%)	3 (1%)
Diarrhoea	44 (18%)	40 (17%)	48 (20%)	11 (5%)
Vomiting	19 (8%)	20 (8%)	29 (12%)	4 (2%)
Constipation	19 (8%)	20 (8%)	25 (10%)	2 (1%)
Dyspepsia	15 (6%)	12 (5%)	25 (10%)	2 (1%)
Decreased appetite	12 (5%)	8 (3%)	20 (8%)	0
Abdominal distension	12 (5%)	12 (5%)	19 (8%)	3 (1%)
Eructation	13 (5%)	10 (4%)	12 (5%)	0
Gastro-oesophageal reflux disease	7 (3%)	6 (2%)	13 (5%)	1 (<1%)
Lipase increased	12 (5%)	14 (6%)	13 (5%)	7 (3%)
Nasopharyngitis	14 (6%)	13 (5%)	16 (7%)	10 (4%)
Diabetic retinopathy	5 (2%)	8 (3%)	8 (3%)	17 (7%)
Vulvovaginal mycotic infection†	0	0	1 (1%)	6 (5%)
Hyperglycaemia	14 (6%)	6 (2%)	3 (1%)	15 (6%)
All gastrointestinal adverse events	112 (47%)	112 (46%)	130 (54%)	29 (12%)
Other adverse events of interest				
Hypoglycaemia (blood glucose <54 mg/dL)	1 (<1%)	3 (1%)	3 (1%)	3 (1%)
Severe hypoglycaemia‡	0	0	0	0
Initiation of rescue therapy for severe persistent hyperglycaemia	9 (4%)	4 (2%)	2 (1%)	13 (5%)
Diabetic retinopathy complications, diabetic macular oedema, and related complications	7 (3%)	11 (5%)	10 (4%)	18 (8%)
Adjudication-confirmed pancreatitis§	1 (<1%)	1 (<1%)	0	0
Treatment-emergent gallbladder and biliary tract disorders§	3 (1%)	1 (<1%)	1 (<1%)	2 (1%)
Cholelithiasis	2 (1%)	1 (<1%)	0	1 (<1%)
Adjudication-confirmed MACE¶	1 (<1%)	1 (<1%)	2 (1%)	2 (1%)
Urinary tract infection	7 (3%)	3 (1%)	5 (2%)	11 (5%)

Data are n (%) of participants for the safety datapoints set. Participants can be counted in more than one category. MACE=major adverse cardiac event. TEAE=treatment-emergent adverse event. *Deaths are also included as serious adverse events and discontinuations due to adverse events. †Denominator adjusted because this is a sex-specific event for females: n=117 (orforglipron 3 mg); n=121 (orforglipron 12 mg); n=123 (orforglipron 36 mg); n=113 (dapagliflozin 10 mg). ‡Severe hypoglycaemia is characterised by altered mental and/or physical status requiring assistance for treatment of hypoglycaemia. Numbers for hypoglycaemia exclude hypoglycaemic events occurring after initiation of any new glucose-lowering agent. §One participant in the orforglipron 3 mg group reported gallstone pancreatitis, which was included in both adverse event categories of "adjudication-confirmed pancreatitis" and "treatment-emergent gallbladder and biliary tract disorders". ¶IMACE (confirmed by clinical endpoint committee) includes: deaths due to cardiovascular cause, myocardial infarction, hospitalisation for unstable angina or heart failure, coronary interventions (such as coronary artery bypass graft or percutaneous coronary intervention), and cerebrovascular events, including cerebrovascular accident (stroke) and transient ischaemic attack.

Table 2: Adverse events and safety during the study period

changes from baseline in DBP for all doses are presented in figure 3E and the appendix (pp 25–28) for the treatment regimen estimand, and in the appendix (pp 29–33) for the efficacy estimand.

Improvements from baseline to week 40 were seen in patient-reported outcomes including all SF-36 v2 component summary scores in all treatment groups (efficacy estimand; appendix p 34), except for mental component and role-emotional scores in the orforglipron 3 mg, orforglipron 12 mg, and dapagliflozin 10 mg groups; as well as vitality and social functioning scores in the dapagliflozin 10 mg group. At week 40, similar changes in DTSQc components were observed across all treatment groups including perceived frequencies of hypoglycaemia and hyperglycaemia, and treatment satisfaction score (efficacy estimand; appendix p 34).

Overall, TEAEs were reported by 632 (66%) of 962 participants. More participants receiving orforglipron reported TEAEs than with dapagliflozin, including 165 (69%) of 240 participants with orforglipron 3 mg, 168 (70%) of 241 with orforglipron 12 mg, and 170 (71%) of 241 with orforglipron 36 mg compared with 129 (54%) of 240 with dapagliflozin 10 mg (table 2). A full TEAE profile with actual numbers of events for each treatment group can be found in the appendix (pp 35–36).

Reports of SAEs were similar across all treatment groups, being observed in eight (3%) of 240 participants treated with orforglipron 3 mg, 14 (6%) of 241 treated with orforglipron 12 mg, nine (4%) of 241 treated with orforglipron 36 mg, and 11 (5%) of 240 treated with dapagliflozin 10 mg (table 2). Among adverse events of special interest, reports of level 2 hypoglycaemia (blood glucose <54 mg/dL) were low and comparable among all treatment groups (one [<1%] of 240, three [1%] of 241, and three [1%] of 241 with orforglipron 3 mg, 12 mg, and 36 mg, respectively, and three [1%] of 240 with dapagliflozin). No case of severe hypoglycaemia was reported. Initiation of rescue therapy for severe persistent hyperglycaemia was reported in nine (4%) of 240, four (2%) of 241, and two (1%) of 241 participants receiving orforglipron 3 mg, 12 mg, and 36 mg, respectively, versus 13 (5%) of 240 receiving dapagliflozin. Diabetic retinopathy or related complications were reported for seven (3%) of 240, 11 (5%) of 241, and ten (4%) of 241 participants in the orforglipron 3 mg, 12 mg, and 36 mg groups, respectively, and 18 (8%) of 240 participants treated with dapagliflozin (table 2). Based on fundus photography evaluation, retinopathy was diagnosed in 110 (11%) of 962 total participants at baseline and rates were balanced across all treatment groups (appendix p 37). Among the participants assessed for retinopathy post-baseline, 40 (5%) of 835 experienced worsening of diabetic retinopathy in at least one eye; rates were balanced across all treatment groups. Among the subset with diabetic retinopathy at baseline, 14 (14%) of

99 participants experienced worsening of diabetic retinopathy in at least one eye at week 40, including one (4%) of 28, five (18%) of 28, and six (30%) of 20 participants receiving orforglipron 3 mg, 12 mg, and 36 mg, respectively, and two (9%) of 23 participants receiving dapagliflozin (appendix p 37). A total of 26 (3%) of 962 participants reported a case of urinary tract infection, including 11 (5%) of 240 among the dapagliflozin treatment group (table 2). Vulvovaginal mycotic infection was reported by one (<1%) of 361 total female participant receiving orforglipron, and by six (5%) of 113 female participants receiving dapagliflozin (table 2).

The most frequent adverse events were of gastrointestinal nature and occurred more frequently with orforglipron 3 mg (112 [47%] of 240), 12 mg (112 [46%] of 241), and 36 mg (130 [54%] of 241) than with dapagliflozin (29 [12%] of 240; table 2). Nausea, diarrhoea, vomiting, and constipation were the most frequently reported gastrointestinal adverse events across orforglipron treatment groups, were typically transient, generally mild-to-moderate in severity, and occurred predominantly during the dose escalation period, with rates attenuating thereafter (table 2, appendix pp 38–39, 55). Intervention discontinuations due to gastrointestinal adverse events were higher in the orforglipron groups (3 mg: 14 [6%] of 240; 12 mg: 18 [7%] of 241; 36 mg: 20 [8%] of 241) than with dapagliflozin (one [<1%] of 240), with a dose-dependent trend across orforglipron groups (appendix p 40). Study intervention discontinuation before reaching randomised dose due to gastrointestinal adverse events occurred in four (2%) of 240, 11 (5%) of 241, and 14 (6%) of 241 participants receiving orforglipron 3 mg, 12 mg, and 36 mg, respectively (appendix p 40).

Changes in pancreatic amylase and serum lipase are shown in the appendix (pp 41–42). Two cases of adjudication-confirmed pancreatitis occurred in orforglipron treatment groups (table 2); one case in the orforglipron 3 mg group was reported as gallstone pancreatitis, and one case in the orforglipron 12 mg group was reported as chronic recurrent pancreatitis. Treatment-emergent gallbladder and biliary tract disorders were reported by three (1%) of 240 participants receiving orforglipron 3 mg (including the case of gallstone pancreatitis), one (<1%) of 241 receiving orforglipron 12 mg, and one (<1%) of 241 receiving orforglipron 36 mg, and by two (1%) of 240 participants receiving dapagliflozin (table 2). Cases of cholelithiasis were reported by two (1%) of 240, one (<1%) of 241, and zero of 241 participants receiving orforglipron 3 mg, 12 mg, and 36 mg, respectively, and by one (<1%) of 240 participants receiving dapagliflozin (table 2). Cases of adjudication-confirmed MACE were reported by one (<1%) of 240, one (<1%) of 241, and two (1%) of 241 participants receiving orforglipron 3 mg, 12 mg, and 36 mg, respectively, and by two (1%) of 240 participants

receiving dapagliflozin (table 2). There were no cases of thyroid malignancy. Changes in ALT and AST are shown in the appendix (pp 41–42). Among the 14 cases of baseline ALT greater than three times the ULN, and the five cases of baseline AST greater than three times the ULN, one participant in the orforglipron 3 mg group shifted to a greater ALT category (≥ 5 to <10 times the ULN) post-baseline, and no shift to a higher category was observed for AST (appendix pp 43–45). One participant had post-baseline ALT of at least ten times the ULN in the orforglipron 3 mg group, one participant had post-baseline AST of at least ten times the ULN in the orforglipron 36 mg group, and one participant had both post-baseline ALT and AST of at least ten times the ULN in the orforglipron 36 mg group; narratives for these three participants are provided in the appendix (pp 44–45). No case met Hy's law criteria for hepatocellular drug-induced liver injury.

At week 40, mean change from baseline in pulse rate was 1.4 beats per min (SE 0.5) with orforglipron 3 mg, 2.7 beats per min (0.6) with orforglipron 12 mg, 4.1 beats per min (0.6) with orforglipron 36 mg, and -0.9 beats per min (0.5) with dapagliflozin (appendix pp 41–42). At safety follow-up, mean pulse rate had returned to near-baseline value for all treatment groups (appendix p 56).

One death (male participant, 43 years old) occurred in the dapagliflozin 10 mg group during the study (71 days from first study dose; 5 days since last study dose), which was adjudicated by an external committee as a non-cardiovascular death due to infection (septic shock), and was not considered to be related to treatment.

Discussion

ACHIEVE-2 is the first head-to-head study to evaluate the efficacy and safety of a non-peptide GLP-1 receptor agonist with an SGLT2 inhibitor in participants with type 2 diabetes and inadequate glycaemic control with metformin. All evaluated doses of orforglipron (3 mg, 12 mg, and 36 mg) demonstrated non-inferiority versus dapagliflozin 10 mg in lowering HbA_{1c} at week 40. Furthermore, superiority was demonstrated across all key secondary endpoints. Beyond HbA_{1c}, orforglipron produced consistent improvements across multiple cardiometabolic parameters, including FSG, bodyweight, BMI, waist circumference, non-HDL cholesterol, triglycerides, and SBP, suggesting broad metabolic benefits. The superiority of orforglipron versus dapagliflozin in reducing HbA_{1c} (up to -0.75% of ETD at week 40) is consistent with previous findings from head-to-head trials of GLP-1 receptor agonists and SGLT2 inhibitors: in the PIONEER-2 trial, oral semaglutide 14 mg demonstrated superiority to empagliflozin 25 mg, another SGLT2 inhibitor, in HbA_{1c} reduction (up to -0.4% of ETD at week 26), in a similar population.¹⁵ Furthermore, the mean change from baseline in HbA_{1c} of -1.56% observed with orforglipron 36 mg at week 40

was consistent with the ACHIEVE-3 study also with orforglipron 36 mg in people with type 2 diabetes inadequately controlled with metformin (up to -1.71% after 52 weeks).¹⁰

These findings build on the established efficacy of orforglipron monotherapy in ACHIEVE-1,⁹ and the demonstrated non-inferiority and superiority of orforglipron 12 mg and 36 mg to oral semaglutide 7 mg and 14 mg at week 52, in combination with metformin, in ACHIEVE-3.¹⁰ Extending this evidence, ACHIEVE-2 provides head-to-head data comparing orforglipron with an SGLT2 inhibitor, a drug class recommended alongside GLP-1 receptor agonists as second-line therapy for glycaemic management in current guidelines.^{5,6}

Current clinical guidelines for the management of type 2 diabetes emphasise the need for effective strategies to manage bodyweight, with weight reduction of 5% to 10% conferring metabolic improvements.⁵ Orforglipron resulted in significantly greater reductions in bodyweight versus dapagliflozin of up to -6.8% (-6.3 kg), with up to 59% of participants having weight reduction of 5%, and up to 28% of participants having weight reduction of 10%. These results are consistent with findings from the placebo-controlled ATTAIN-2 study, where orforglipron demonstrated superior bodyweight change of up to -9.6% at week 72 in adults with overweight or obesity with type 2 diabetes.¹¹ Orforglipron 3 mg resulted in greater HbA_{1c} reduction than dapagliflozin while providing comparable weight reduction. This could suggest that clinically relevant glycaemic control is achievable irrespective of modest weight reduction.

Dapagliflozin has demonstrated cardiovascular risk reduction in dedicated outcomes trials such as DECLARE-TIMI, DAPA-HF, and DAPA-CKD,^{12,18,19} highlighting its relevance as a comparator with proven cardiorenal benefits. However, ACHIEVE-2 was not designed or powered to assess cardiovascular outcomes, and no conclusions regarding comparative cardiovascular efficacy can be drawn from these data. However, the significant improvements in cardiometabolic risk markers including SBP, non-HDL cholesterol, and triglycerides with orforglipron 36 mg compared with dapagliflozin highlight the importance of further evaluating orforglipron in dedicated cardiovascular trials, including the ACHIEVE-4 (NCT05803421) safety trial, and the ATTAIN-OUTCOMES (NCT07241390) cardiovascular outcomes trial, which will definitively assess whether these surrogate markers translate to cardiovascular risk reduction.

The overall incidence of adverse events was greater with orforglipron treatment than with dapagliflozin; however, no unexpected safety signals emerged throughout the study duration. The most frequent adverse events associated with orforglipron were gastrointestinal adverse events, including nausea, vomiting, diarrhoea, and constipation. These effects

were mostly mild-to-moderate in severity and occurred mainly during dose escalation. These findings align with safety outcomes reported in other orforglipron studies,^{9,11,20} as well as other GLP-1 receptor agonists.^{21,22} The higher incidence of urinary tract infection and vulvovaginal infections with dapagliflozin was expected and similar safety findings have been reported previously with the SGLT2 inhibitor class.^{12,23} More study intervention discontinuations were observed in the orforglipron groups. Gastrointestinal adverse events were the main cause of discontinuation in orforglipron treatment groups. While study intervention discontinuations tended to be dose-dependent, orforglipron 12 mg and 36 mg had similar rates of discontinuation after the first two doses during escalation. Although few cases were observed, the risk of worsening of diabetic retinopathy highlights the importance of screening in practice, and that retinopathy should be closely monitored in people with type 2 diabetes.⁶ The frequency of SAEs remained low across all treatment groups (4% overall, 42 of 962). Overall, treatment with orforglipron was associated with increased gastrointestinal adverse events and increased discontinuations due to gastrointestinal adverse events compared with dapagliflozin, whereas rates of SAEs, hypoglycaemia, and other adverse events of interest were low and balanced across treatment groups. We acknowledge that in the current context of a clinical study, the forced dose-escalation scheme might have resulted in adverse event rates that would differ from a real-world setting. In clinical practice, treatment schemes would be tailored to patient individual needs and goals based on an individualised patient-centred approach. For instance, patients might change drug dosing based on disease progression and tolerability.

Regarding the hepatic safety of orforglipron, participants with MASLD were permitted to enrol provided that ALT or AST concentrations were not more than five times the ULN, reflecting the high prevalence of MASLD in people living with type 2 diabetes and ensuring a generalisable study population. Across ACHIEVE-2 there was no evidence of dose-dependent transaminase increases and no cases of Hy's law. Among the 17 participants with elevated baseline transaminases between three and five times the ULN, 16 participants remained stable or demonstrated improvements. The two reports of transient transaminase elevations to at least ten times the ULN while taking orforglipron occurred in the setting of confounding aetiologies. A third participant experienced a transaminase elevation to at least ten times the ULN, which occurred approximately 5 months after discontinuation of orforglipron concurrently with a creatine kinase elevation of 76 times the ULN, suggesting an alternative cause.

The ACHIEVE-2 study had several limitations. Although orforglipron dose was blinded, the open-label

design might have introduced bias in adverse event reporting and potentially influenced adherence behaviours, which could affect both efficacy and tolerability assessments. Another limitation is that gastrointestinal adverse events were self-reported by participants, which might also bias data collection and interpretation. Additionally, this study reports clinical endpoints that have not been tested for multiplicity (type I error rate) from which cautious interpretations should be drawn.²⁴ Finally, ACHIEVE-2 enrolled participants across six countries in Asia, Europe, and North America. The racial and ethnic composition of the study population largely reflected the demographics of the enrolling regions, which might not be representative of the global population of individuals living with type 2 diabetes.

A key strength of this study is the high study completion rate, resulting in minimal missing data. The global footprint of the trial, as well as the balance between male and female participants, improves the generalisability of the results. Furthermore, the use of an active comparator with proven glycaemic, cardiovascular, and renal benefit improves the clinical relevance of the results. Finally, the oral formulation, absence of fasting requirements, and room temperature stability of orforglipron could address practical barriers that impact initiation of peptide-based injectable or oral GLP-1 receptor agonists.

In conclusion, in the ACHIEVE-2 study, orforglipron at once-daily doses ranging from 3 mg to 36 mg was non-inferior and superior to dapagliflozin 10 mg with clinically meaningful reductions in HbA_{1c}.^{25–27} Orforglipron doses of 12 mg and 36 mg also demonstrated superiority to dapagliflozin 10 mg in bodyweight reduction, and orforglipron 36 mg was superior to dapagliflozin 10 mg in decreasing key cardiometabolic risk factors including triglycerides, non-HDL cholesterol, and SBP. Orforglipron demonstrated a safety and tolerability profile consistent with that of the GLP-1 receptor agonist class, mainly characterised by increased rates of gastrointestinal adverse events and intervention discontinuations. These findings support orforglipron as a potential therapeutic option for type 2 diabetes management in combination with metformin, although cardiovascular and renal outcome assessments will be necessary to inform a potential implementation in the treatment algorithm.

Contributors

The sponsor (Eli Lilly and Company) conceived, designed, and supervised the study, which included site monitoring, data collection, and analysis. YC, MD, W-SW, and PODP contributed to the study design. YC and MD provided medical oversight during the study. MW, MD, PODP, and YC accessed and verified the data. PODP, JL, and RL were responsible for the data curation, formal analysis, validation, visualisation, and editing of statistical analyses. YC, MD, and PODP drafted the manuscript. MW, TF, WJ, W-SW, JL, RL, and ME participated in the critical review of the manuscript. All authors participated in the interpretation of the data and approved the final and

previous versions of the manuscript, had full access to all the data in the study, and had final responsibility for the decision to submit for publication.

Declaration of interests

MW declares payment as a member of a speaker bureau, support for attending meetings and/or travel, and stocks from Eli Lilly and Company. TF declares consulting fees from AstraZeneca, Atrogi, Bayer, Cipla, Eli Lilly and Company, Eysense, Fortbildungskolleg, Novo Nordisk, Pfizer, Sanofi, Remynd, and Roche; payments for speaker services to Amarin, AstraZeneca, Böhlinger Ingelheim, Berlin Chemie, Cipla, Daiichi Sankyo, Eli Lilly and Company, Fortbildungskolleg, MSD, Novartis, Novo Nordisk, Sanofi, and Santis; and support for attending meetings and/or travel from AstraZeneca and Eli Lilly and Company. WJ declares unpaid roles as International Diabetes Federation Western Pacific Region chair-elect, and Director of the Asian Association for the Study of Diabetes. PODP, MD, W:SW, RL, JL, ME, and YC are employees and shareholders of Eli Lilly and Company.

Data sharing

Lilly provides access to all individual participant data collected during the study, after anonymisation, with the exception of pharmacokinetic or genetic data. Data are available to request 6 months after the indication studied has been approved in the USA and EU and after primary publication acceptance, whichever is later. No expiration date of data requests is currently set once data are made available. Access is provided after a proposal has been approved by an independent review committee identified for this purpose and after receipt of a signed data sharing agreement. Data and documents, including the study protocol, statistical analysis plan, clinical study report, and blank or annotated case report forms, will be provided in a secure data sharing environment. For details on submitting a request, see the instructions provided at www.vivli.org.

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